SOHAM VYAS

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SUMMARY

AI Engineer with 5+ years of experience building and deploying production AI/ML systems across FinTech, Healthcare, and Clean Energy. Experienced in Generative AI, LLM applications, DocumentAI, RAG, Computer Vision, Speech-to-Text, scalable ML pipelines, and AI inference systems. Proven track record of architecting production workflows processing 50K+ documents/month, improving AI operating efficiency by 30%, and delivering end-to-end AI solutions from model development through deployment and enterprise integration. Inventor with granted and published patents and 9+ research publications.

CORE EXPERTISE

Generative AI	LLMs, RAG, Prompt Engineering, Fine-tuning, LoRA/QLoRA, vLLM, LangChain, LangGraph, AI Agents, Vector Databases, LLMOps
AI/ML	PyTorch, TensorFlow, Keras, HuggingFace, Scikit-learn, ONNX Runtime, Computer Vision, NLP, OCR, Speech-to-Text, NER
AI Systems	Scalable ML Pipelines, Async Processing, FastAPI, Flask, REST APIs, Docker, Kubernetes, CI/CD, MLflow, Triton Inference Server
Cloud	Azure AI, Azure OpenAI, GCP, Apache Airflow, Linux, Git, Python Automation
Inference	vLLM, ONNX, Quantization, Low-Latency Inference, EdgeAI, Real-Time Pipelines
AI Domains	DocumentAI, FinTech AI, Healthcare AI, Computer Vision, Medical AI, Clean Energy
Model Architectures	Transformers, CNNs, RNNs, LSTMs, GANs, Autoencoders

PROFESSIONAL EXPERIENCE

Senior AI Engineer September 2025 – Present
AiVantage Global *Ahmedabad, India*

- Architected and deployed scalable asynchronous DocumentAI systems processing **50,000+ documents/month**, using Azure Service Bus orchestration, soft tenant isolation, and AKS-ready microservices.
- Designed a **LLM-powered Signature Intelligence engine** for contextual document understanding, signature validation, rule-based verification, and integration with legacy banking workflows.
- Built high-throughput **NER and document processing pipelines** for bulk entity extraction, attribute mapping, and multi-source document ingestion across enterprise workflows.
- Engineered concurrent AI processing workflows handling **hundreds of documents simultaneously** with production throughput exceeding **1,000+ enterprise documents/day**.
- Developed observability and AI cost-tracking mechanisms across processing stages, contributing to a **30% reduction in AI operating costs** while supporting scalable multi-tenant processing.
- Owned end-to-end AI application delivery spanning architecture, backend engineering, model integration, orchestration, deployment optimization, and enterprise workflow integration using **LLMs, RAG, LLMOps, and Azure AI**.
- Contributed to a fintech AI platform nominated for the **Bank Tech Awards 2026** under the "Best AI Solution" category.

AI/ML Engineer October 2024 – August 2025
Artem HealthTech *Ahmedabad, India*

- Developed LLM-powered **Clinical Decision Support System** applications for EHR question answering and healthcare intelligence workflows.
- Built multilingual Indian **Speech-to-Text and OCR pipelines** achieving 90%+ accuracy and processing **30,000+ laboratory reports/month** with automated LOINC code mapping.

Adhoc Faculty, Computer Science July 2023 – September 2024
Pandit Deendayal Energy University *Gandhinagar, India*

- Taught Computer Science and guided research teams working across **Green Energy, Environment, and Computer Vision**.
- Led and mentored multiple research teams through project design, experimentation, implementation, and evaluation.

AI Intern May 2022 – May 2023
Larsen & Toubro Technology Services (LTTS) *Vadodara, India*

- Developed AI application prototypes for **Metaverse platforms** using Blender Game Engine and computer vision models.
- Built Python automation to integrate vision-based AI models into runtime client-server architectures.

Assistant System Engineer – Trainee January 2020 – January 2021
Tata Consultancy Services (TCS) *Ahmedabad, India*

- Supported SRE operations for client applications in an Agile engineering environment across Linux and cloud infrastructure.
- Maintained **5 production applications** using Linux, GCP, Apache Airflow, and Python automation workflows.

AI PROJECTS

AI in Clean Energy 2021 – 2024

- Developed forecasting, predictive maintenance, and battery aging models using real-world power plant data from PDEU/GERMI, achieving up to **96% prediction accuracy**.
- Built impact analysis and maintenance activity prediction pipelines validated on live power plant environments, achieving **89% prediction accuracy**.

Computer Vision for Autonomous Surveillance 2023 – 2024

- Developed computer vision pipelines for autonomous surveillance vehicles, including obstacle avoidance and dynamic path remapping.
- Developed technology underlying a published patent for automatic mapping of unmanned surveillance vehicles.

EDUCATION

M.Tech in Data Science, Pandit Deendayal Energy University 2021 – 2023
CGPA: **9.84/10** – Gold Medalist

B.E. in Computer Engineering, LDRP-ITR 2015 – 2019
CGPA: 7.85/10

PATENTS

Device for Solar Energy Prediction with Maintenance and Weather Analysis 2025
Granted Design Patent, Indian Patent Registry No. 415558-001

Modular Device for Multi-Terrain Autonomous Driving for Unmanned Vehicles 2023
Published Indian Patent, Indian Patent Registry No. 202321038021

SELECTED PUBLICATIONS

9+ publications across AI, Computer Vision, Clean Energy, and Explainable AI.

The Impacts of Maintenance Weather and Aging on Solar Power Generation Forecasting and Prediction 2023
Presented at IEEE IAS GlobConHT, Maldives

Prediction of Solar Power Generation and Maintenance Activities for 1MW Power Plant 2023
Presented at IEEE IAS GlobConET, London

IoT Based Soyabean Pest Classification Using Transfer Learning and Explainable AI 2024
Presented at IEEE CONECCT 2024, IISc Bengaluru